CS221-EndSem-Part-C-30Minutes

CS221-EndSem-Part-C

...

Points: 34/40

1

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✓ **Correct** 1/1 Points

4

FPGA contents

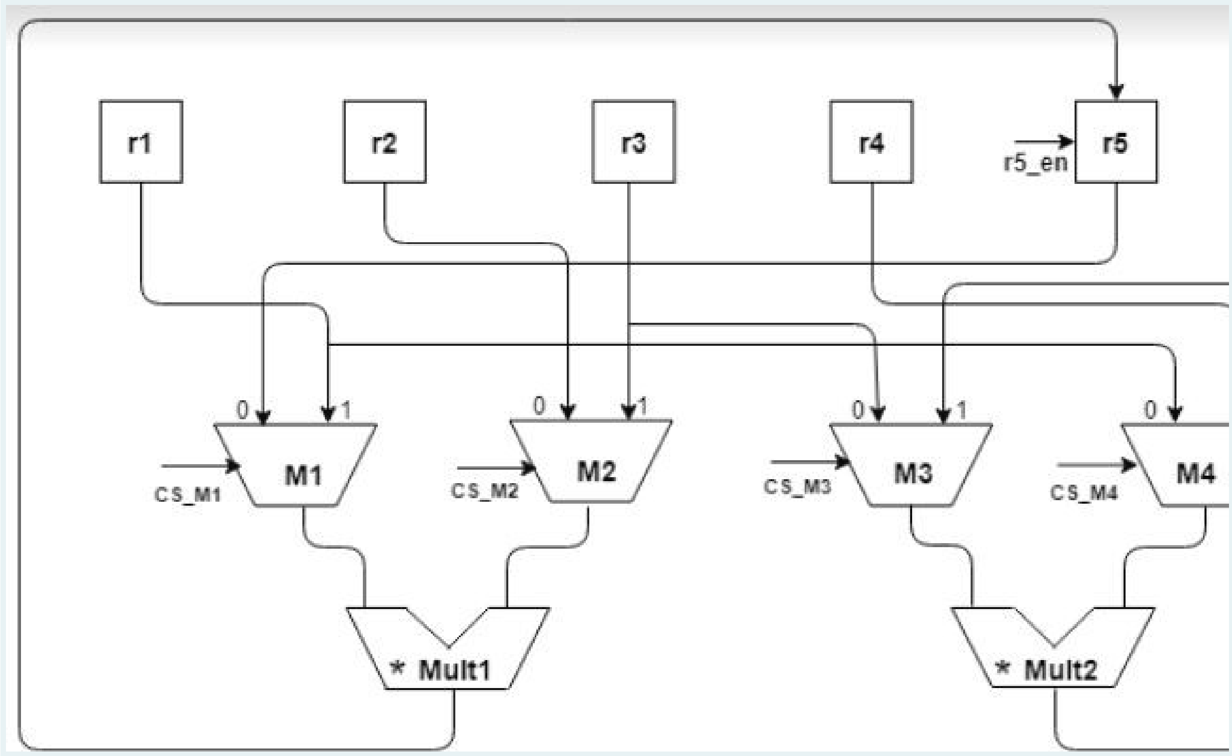
- Configurable Logic blocks
- Configurable I/O blocks
- Configurable Interconnect
- All of the above ✓

✓ **Correct** 2/2 Points

5

Considering the datapath in Figure. M1, M2, M3 and M4 blocks are multiplexor block, Mult1 and Mult2 are multiplier block. Assume vertical-horizontal crossings as not connected but tri-points are connected. Multiplication takes only one cycle

What would be the operations executed in the datapath, if the control signal assignment is (1, 1, 0, 0, 1, 1), where the order of control signals is CS_M1, CS_M2, CS_M3, CS_M4, r5_en, r6_en)?



- r5 = r1*r2 and r6 = r3*r4
- r5 = r5 * r3 and r6 = r1 * r2
- r5 = r1 * r3 and r6 = r3 * r1 ✓
- r5 = r5 * r2 and r6 = r1 * r2

✓ **Correct** 1/1 Points

6

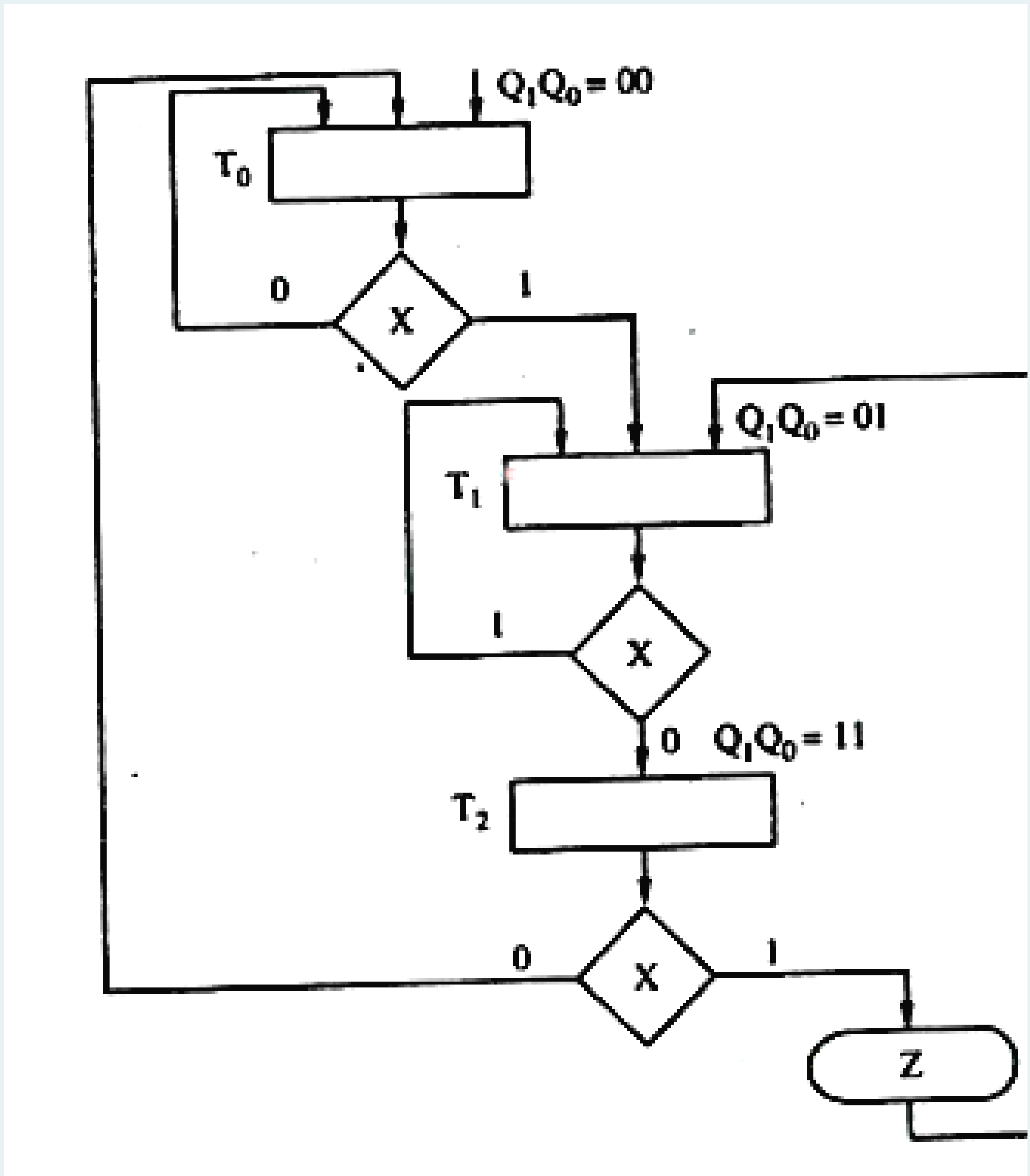
Conditional Box in ASM is also called as

- Decision box
- Transition box ✓
- State box
- Vector decision box

✓ **Correct** 1/1 Points

7

What sequence will be detected by the given ASM?



- 011
- 100
- 101 ✓
- 111

✓ **Correct** 1/1 Points

8

Which of the following statement(s) is/are correct for ASM?
S1: ASM can model Moore type of FSM
S2: ASM can model Mealy type of FSM
S3: ASM can model mixed type of FSM

- Only S1
- Only S2
- Only S3
- All S1, S2 and S3 ✓

✓ **Correct** 2/2 Points

9

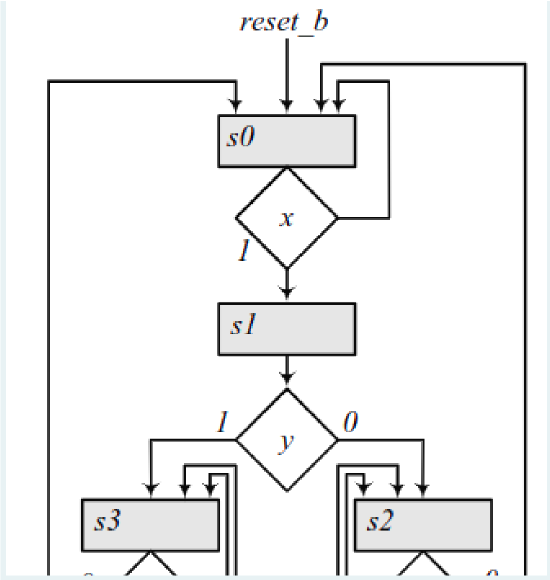
Given a four variable function F(w, x, y, z) needs to be implemented using two 3 input LUTs, choose the correct implemention the same.

- y.F(w, x, 0, z) + y'.F(w, x, 1, z)
- w'.F(w, 0, y, z) + w.F(w, 1, y, z)
- w.F(0, x, y, z) + w'.F(1, x, y, z)
- w'.F(0, x ,y, z) + w.F(1, x, y, z) ✓

✓ **Correct** 2/2 Points

10

What is the total number of ASM blocks in the given ASM?



- 2
- 4 ✓
- 6
- 10

✗ **Incorrect** 0/2 Points

11

Given the Pseudo-code

x=0; for(i=0;i<34; i+3) { x=x+i; }

Assume each addition requires 1 clock cycle and there is only one adder (no additional increment unit) and one comparator unit in the data path. How many clock cycles require to execute this loop?

- 33
- 12
- 23 ✓
- 34

✓ **Correct** 1/1 Points

12

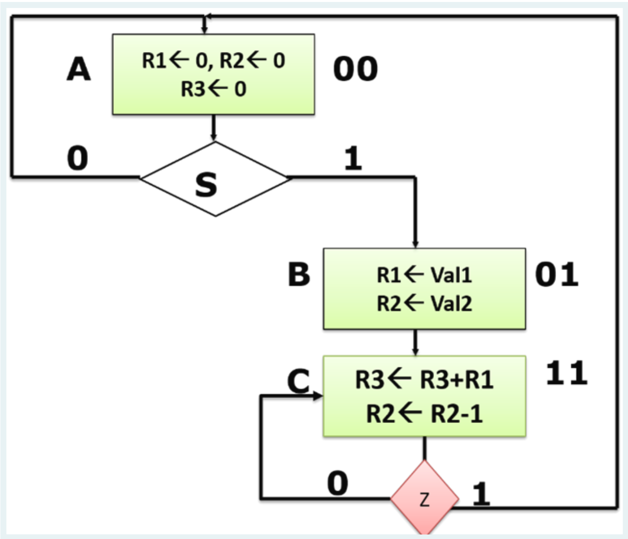
1. Which of the following statement(s) is/are correct in ASM?
S1: state box is of Moore type whereas conditional box is of Mealy type.
S2: state box is of Mealy type whereas conditional box is of Moore type.

- Only S1 ✓
- Only S2
- Both S1 and S2
- Neither S1 and S

✓ **Correct** 2/2 Points

13

Given the bellow ASM, the number of cycles it will take to do one complete operation once S=1



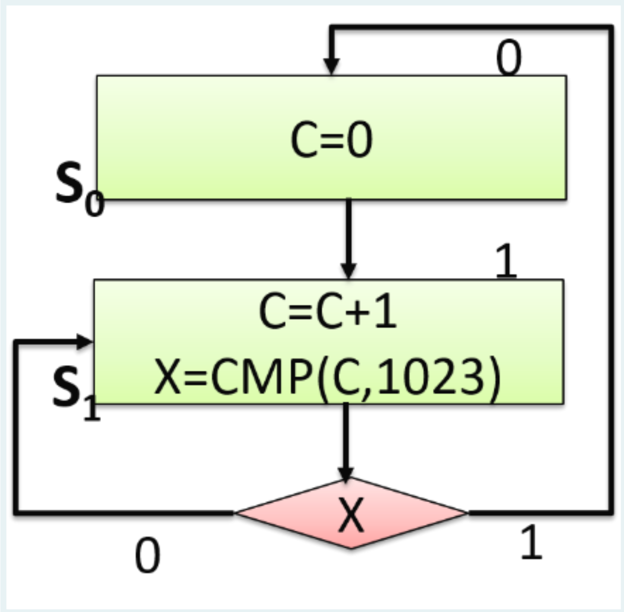
- It does not depend on Val1 and Val2
- It depends on both Val1 and Val2
- It depends on value of Val1
- It depends on value of Val2 ✓

✓ **Correct** 2/2 Points

14

Suppose we want to implement the following ASM using a D-FF and NS logic. The to be generated control signal are CLR. CMP, ADD and LoadReg.

Assume T0 and T1 value are decoded (set to 1) when ASM in state 0 and state 1 respectively.



Select the correct implementation.

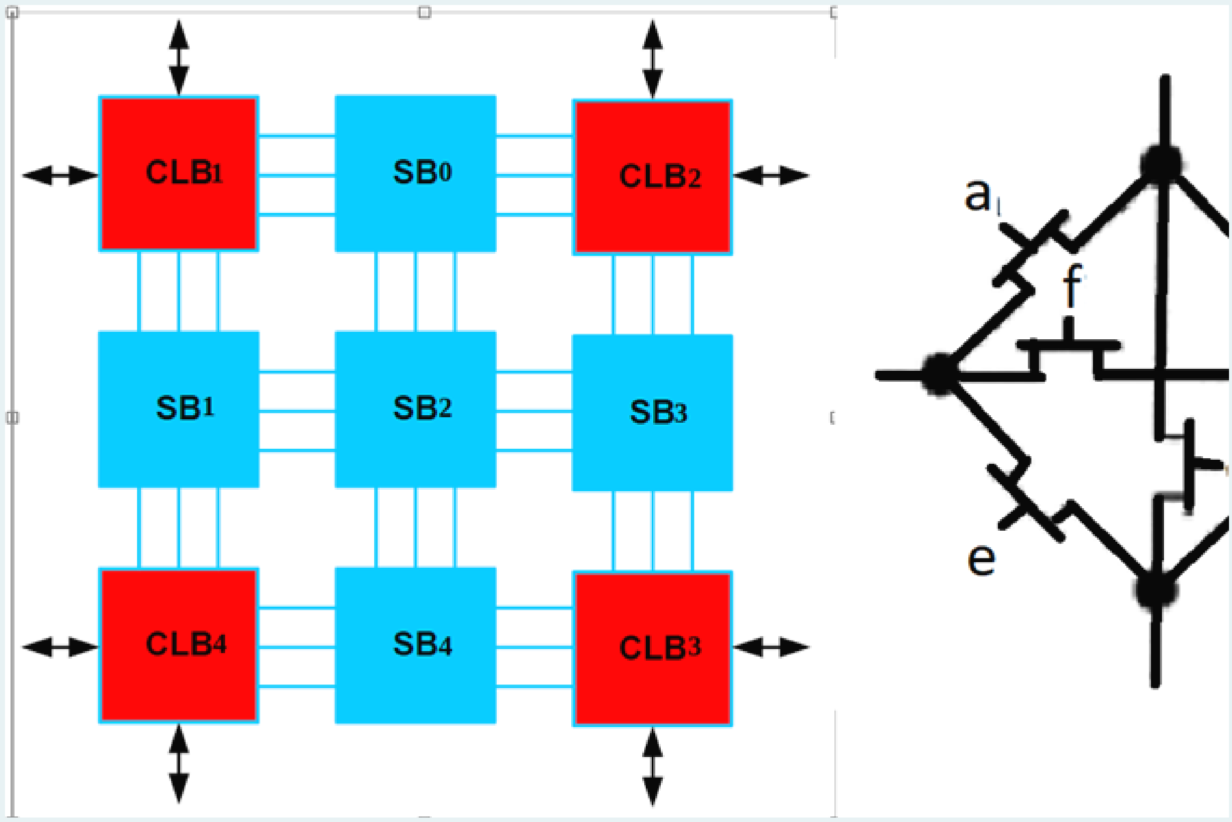
- CLR=T0, CMP=T1, ADD=T1, D=T1.X'+T0, LoadReg=T1'
- CLR=T0', CMP=T1, ADD=T1, D=T1.X'+T0, LoadReg=T1
- CLR=T0, CMP=T1', ADD=T1, D=T1.X'+T0, LoadReg=T1
- CLR=T0, CMP=T1, ADD=T1, D=T1.X'+T0, LoadReg=T1 ✓

✓ **Correct** 2/2 Points

15

Consider the below configuration bits for switch boxes

SB3: (a,b,c,d,e,f) = (0,0,0,1,0,0)
SB4: (a,b,c,d,e,f) = (0,0,0,0,0,1)
Which of the below CLBs are connected with the above configuration?

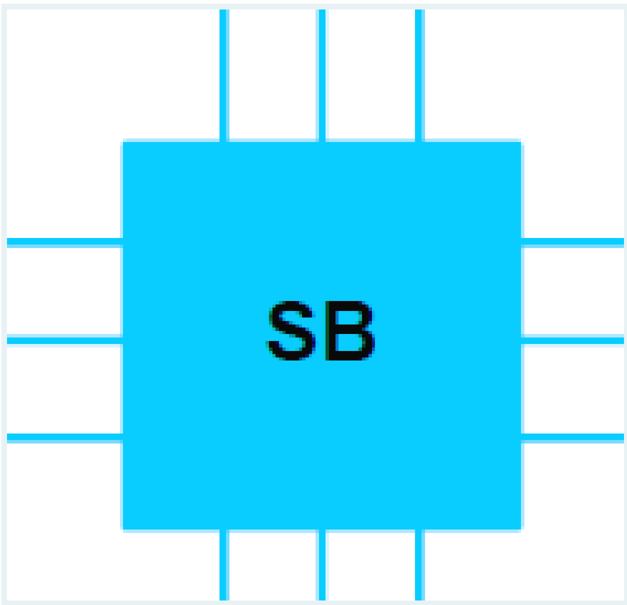


- CLB2 will be connected to CLB1 and CLB3
- CLB1 will be connected to CLB2 and CLB4
- CLB4 will be connected to CLB1 and CLB3
- CLB3 will be connected to CLB2 and CLB4 ✓

✓ **Correct** 2/2 Points

16

What is the total number of configuration bits required for the below configurable switch box of FPGA?



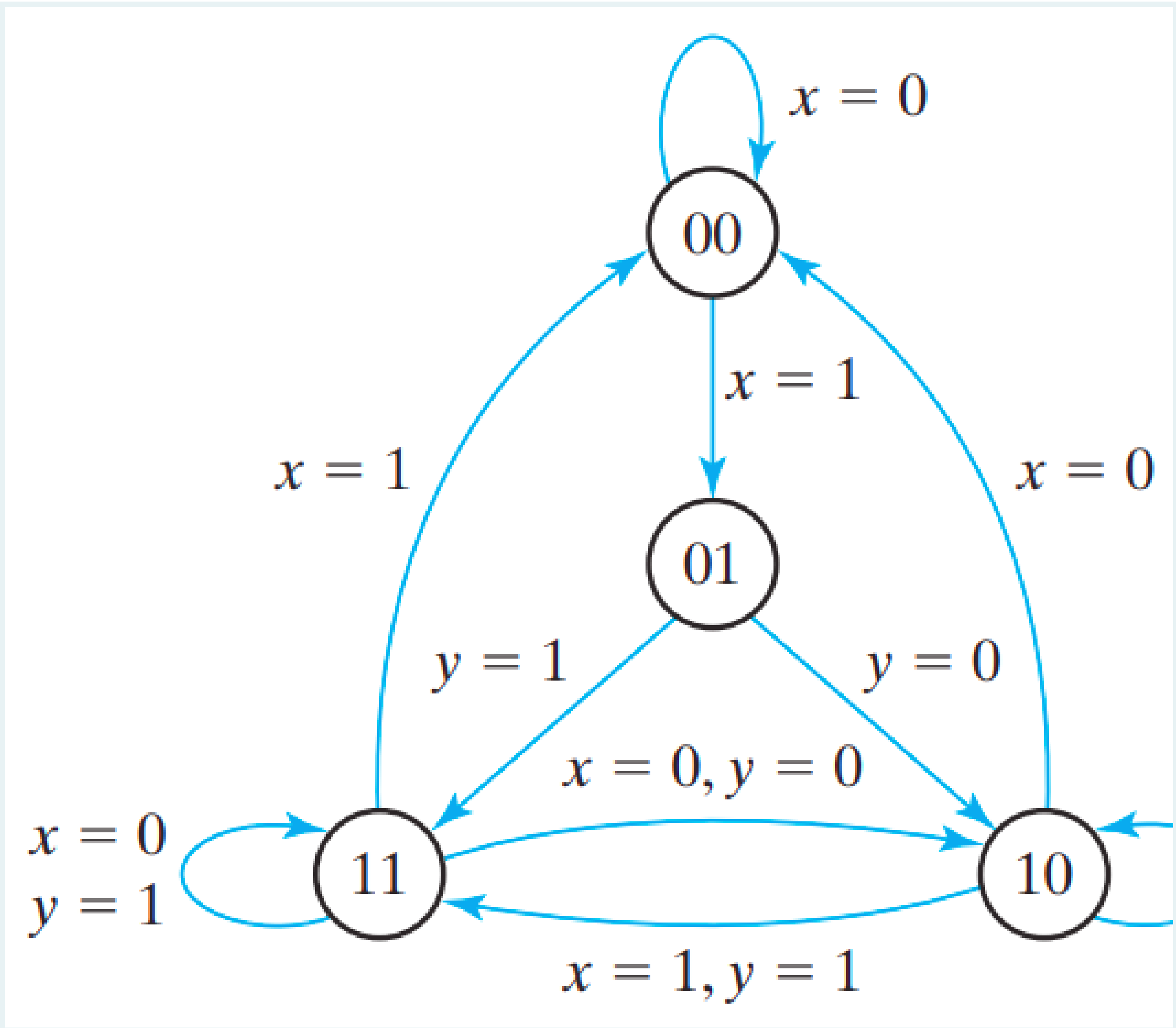
- 6

- 18 ✓
- 20
- 30

✓ **Correct** 2/2 Points

17

What will be minimum number of binary decision boxes present in the corresponding ASM for the below FSM?



- 4
- 5
- 6 ✓

8

Incorrect 0/2 Points

18

What would be the minimum clock cycles required in best case for computation of the following C code in hardware if there is no resource constraint (you can use any/infinite number of adder)?

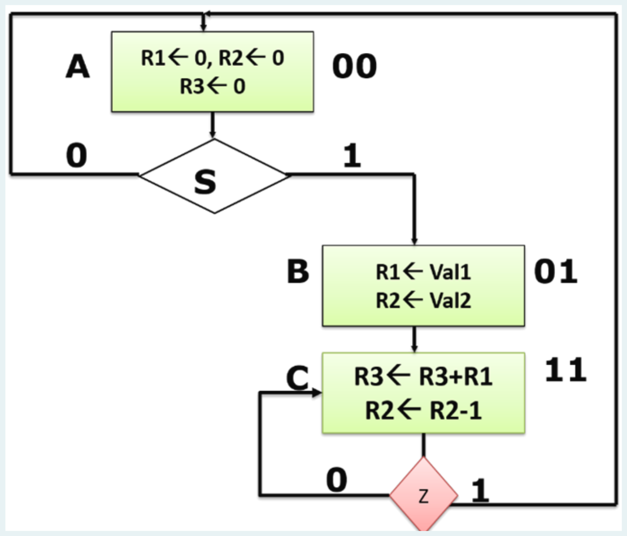
for(i=0; i< 1024; i++) { a = a + b[i]; }

- 1024
- 512
- 32
- 10 ✓

Correct 2/2 Points

19

How many ASM Block will be there in the given ASM

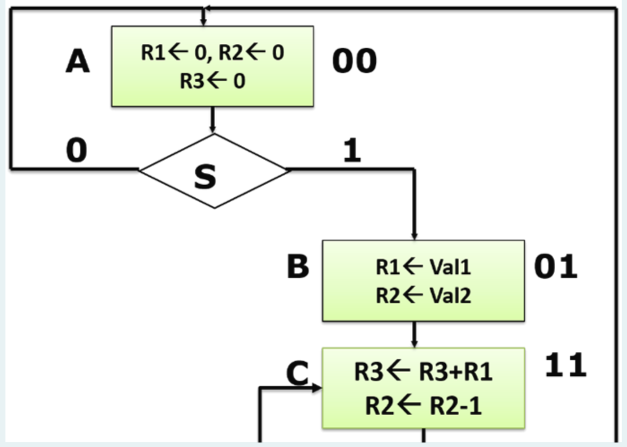


3

Correct 2/2 Points

20

Given the ASM, generate the control signal decr, clears, load12 (load for reg 1 and 2), load3 (load for reg3)



- decr=T2.Z′ clears=T0 load12=T0 load3=T2.Z′
- decr=T2.Z clears=T0 load12=T0 load3=T2.Z
- decr=T2 clears=T0 load12=T0 load3=T2
- None of the above ✓

✓ **Correct** 1/1 Points

21

Which of the following is true?

- All the operations in two ASM blocks can executed in a single clock cycle
- Decision box depends on the status of the current clock cycle
- Decision box depends on operation of current block
- Decision box depends on the status of the previous clock cycle ✓

✗ **Incorrect** 0/2 Points

22

Given the Pseudo-code **for(i=0;i<256;i++) { P=P+Q; R=R+S; }** Assume increment operation can done by register. How many adders needed to complete this operation in 8 cycles?

16

256 ✓

64

512

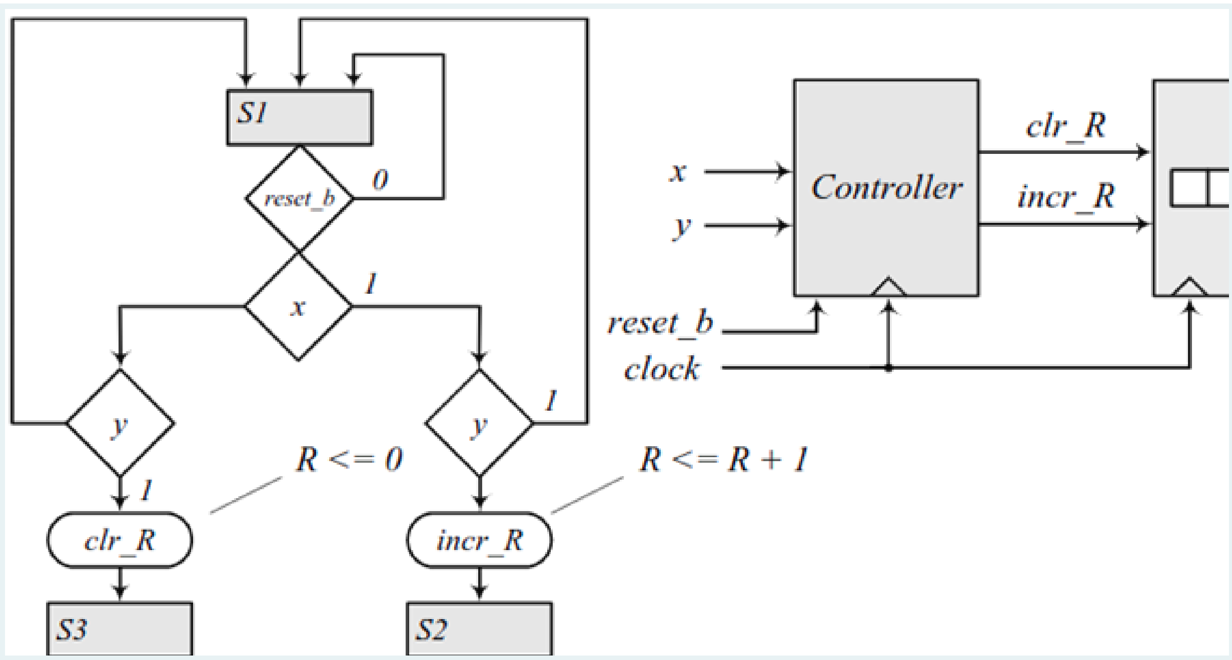
✓ **Correct** 2/2 Points

23

Which of the following statements are correct for the below ASM?

S1: If xy is 01 then R is cleared and control passes from first state to another state.

S2: If xy is 10 then R is incremented by 1 and control passes to another state.



Both S1 and S2 are true ✓

Only S1 is true

Only S2 is true

Both S1 and S2 are false

✓ **Correct** 1/1 Points

24

Assertion (A): ASM do not perform better than FSM for large digital system.

Reason (R): To specify a large digital system having a lot of memory element using state table is difficult because the number of states would be enormous.

- Both (A) and (R) are true and (R) is the correct explanation for (A)
- Both (A) and (R) are true and (R) is not the correct explanation for (A)
- A is true but R is false
- R is true but A is false ✓

✓ **Correct** 1/1 Points

25

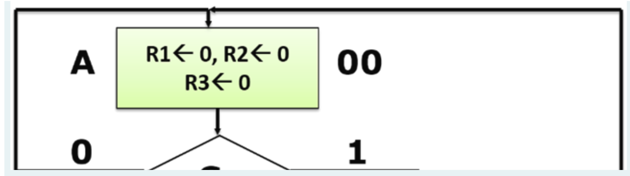
An ASM blocks include

- state box
- decision box
- conditional box
- All the above ✓

✓ **Correct** 2/2 Points

26

For the given ASM, assume state A, B and C represents by decoded signal T0, T1 and T2, what will be value D1 and D0 signal to calculate NS logic



- D1=T1′+T2.Z′ D0= T2+T1.Z′+T0.S
- D1=T1+T2′.Z′ D0= T0.S
- D1=T1+T2.Z′ D0= T1+T2.Z′+T0.S ✓
- D1=T1′+T2.Z′ D0= T1+T2.Z′+T0′.S

✓ **Correct** 2/2 Points

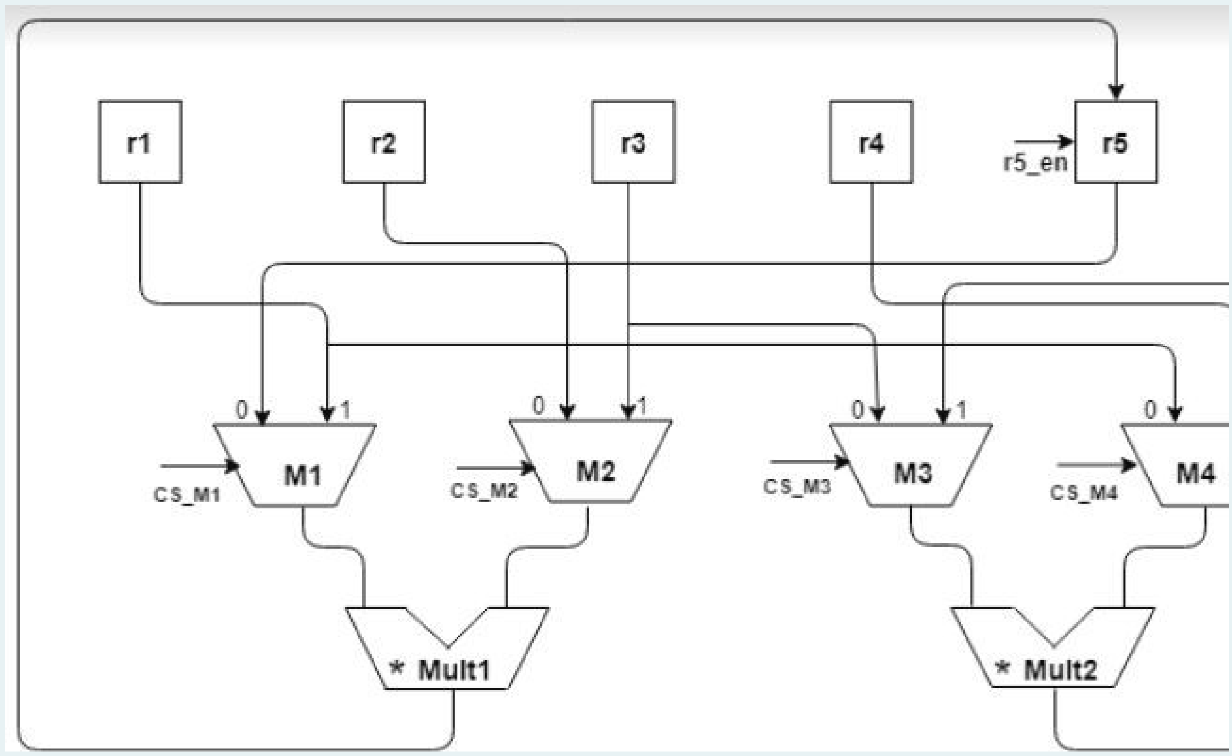
27

Considering the data path in Figure. M1, M2, M3 and M4 blocks are multiplexor blocks. Mult1 and Mult2 are multiplier blocks. Assume vertical-horizontal crossings as not connected but tri-points are connected. Multiplication takes only one cycle. Let us assume that following operations are executing in the data path in time step S1.

r5 = r1 * r2

r6 = r3 * r4

What would be the control signals for the operations in time stamp S1, if the control assertion pattern order is (CS_M1, CS_M2, CS_M3, CS_M4, r5_en, r6_en)?



- (1, 0, 0, 0, 0, 0)
- (1, 0, 0, 1, 1, 1) ✓
- (1, 0, 0, 1, 0, 0)
- (0, 1, 1, 0, 1, 1)